

EXPERIMENT – 4

QUERYING LEDGER DATA WITH HYPERLEDGER FABRIC

Aim

To query and retrieve ledger data from a Hyperledger Fabric blockchain network using JavaScript chaincode, and verify asset information stored in the world state database.

Software Requirements

- Kali Linux / Ubuntu
- Docker
- Docker Compose
- Node.js (Version 18 or later)
- npm
- Git
- Hyperledger Fabric Samples
- Hyperledger Fabric Binaries
- Hyperledger Fabric Docker Images
- Visual Studio Code (Optional)

Hardware Requirements

- Processor: Intel Core i3 or above
- RAM: Minimum 8 GB
- Storage: Minimum 20 GB Free Disk Space
- Internet Connection

Step 1: Navigate to the Fabric Test Network

```
cd ~/fabric-dev/fabric-samples/test-network
```

```
dharsna@DHARSHU:~$ cd ~/fabric-dev/fabric-samples/test-network
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 2: Start the Hyperledger Fabric Network

```
./network.sh up createChannel -ca
```

```
mod_policy : , policies : {}, values : {}, version : 0 , mod_policy : , policies : {},
values":' {},' "version":' "0"' '}}' '}}'
++ configtxlator proto_encode --input /home/dharsna/fabric-dev/fabric-samples/test-network/channel-artifacts/config_update_in_envelope.json --type common.Envelope --output /home/dharsna/fabric-dev/fabric-samples/test-network/channel-artifacts/Org2MSPanchors.tx
2026-09-08 15:30:27.429 UTC 0001 INFO [channelCmd] InitCmdFactory -> Endorser and orderer connections initialized
2026-09-08 15:30:27.440 UTC 0002 INFO [channelCmd] update -> Successfully submitted channel update
Anchor peer set for org 'Org2MSP' on channel 'mychannel'
Channel 'mychannel' joined
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 3: Deploy the Asset Transfer Chaincode

```
./network.sh deployCC \  
-ccl javascript \  
-ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript
```

```
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':  
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vsc, Approvals: [Org1MSP: true, Org2MSP: true]  
Query chaincode definition successful on peer0.org1 on channel 'mychannel'  
Using organization 2  
Querying chaincode definition on peer0.org2 on channel 'mychannel'...  
Attempting to Query committed status on peer0.org2, Retry after 3 seconds.  
+ peer lifecycle chaincode querycommitted --channelID mychannel --name basic  
+ res=0  
Committed chaincode definition for chaincode 'basic' on channel 'mychannel':  
Version: 1.0, Sequence: 1, Endorsement Plugin: escc, Validation Plugin: vsc, Approvals: [Org1MSP: true, Org2MSP: true]  
Query chaincode definition successful on peer0.org2 on channel 'mychannel'  
Chaincode initialization is not required  
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

```
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ export FABRIC_CFG_PATH=$HOME/fabric-dev/fabric-samples/config  
source scripts/envVar.sh  
setGlobals 1  
Using organization 1  
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 4: Initialize the Ledger

```
peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $ORDERER_CA \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $PEER0_ORG1_CA \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $PEER0_ORG2_CA \  
-c '{"function":"InitLedger","Args":[]}'
```

```
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $ORDERER_CA \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $PEER0_ORG1_CA \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $PEER0_ORG2_CA \  
-c '{"function":"InitLedger","Args":[]}'  
2026-09-08 15:33:44.581 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200  
dharshna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 5: Query All Assets

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["GetAllAssets"]}'
```

Observation / Screenshot 5: All initialized asset records displayed successfully

[PASTE SCREENSHOT HERE]

Step 6: Query a Specific Asset

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset1"]}'
```

```
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["GetAllAssets"]}'  
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red",  
"ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":  
10,"docType":"asset"}, {"AppraisedValue":600,"Color":"yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedVal  
ue":700,"Color":"black","ID":"asset5","Owner":"Adriana","Size":15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asse  
t6","Owner":"Michel","Size":15,"docType":"asset"}]  
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 7: Create a New Asset

```
peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $ORDERER_CA \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $PEER0_ORG1_CA \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $PEER0_ORG2_CA \  
-c '{"function":"CreateAsset","Args":["asset20","Green","8","Arun","1500"]}'
```

```
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode invoke \  
-o localhost:7050 \  
--ordererTLSHostnameOverride orderer.example.com \  
--tls \  
--cafile $ORDERER_CA \  
-C mychannel \  
-n basic \  
--peerAddresses localhost:7051 \  
--tlsRootCertFiles $PEER0_ORG1_CA \  
--peerAddresses localhost:9051 \  
--tlsRootCertFiles $PEER0_ORG2_CA \  
-c '{"function":"CreateAsset","Args":["asset20","Green","8","Arun","1500"]}'  
2026-09-08 15:35:06.026 UTC 0001 INFO [chaincodeCmd] chaincodeInvokeOrQuery -> Chaincode invoke successful. result: status:200 payloa  
d:{"ID":"asset20","Color":"Green","Size":8,"Owner":"Arun","AppraisedValue":1500}  
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 8: Query the Newly Created Asset

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset20"]}'
```

```
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset20"]}'  
{"AppraisedValue":1500,"Color":"Green","ID":"asset20","Owner":"Arun","Size":8}  
dharshta@DHARSHU:~/fabric-dev/fabric-samples/test-network$ |
```

Step 9: Query All Assets Again

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["GetAllAssets"]}'
```

```
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["GetAllAssets"]}'  
[{"AppraisedValue":300,"Color":"blue","ID":"asset1","Owner":"Tomoko","Size":5,"docType":"asset"}, {"AppraisedValue":400,"Color":"red",  
"ID":"asset2","Owner":"Brad","Size":5,"docType":"asset"}, {"AppraisedValue":1500,"Color":"Green","ID":"asset20","Owner":"Arun","Size":  
8}, {"AppraisedValue":500,"Color":"green","ID":"asset3","Owner":"Jin Soo","Size":10,"docType":"asset"}, {"AppraisedValue":600,"Color": "  
yellow","ID":"asset4","Owner":"Max","Size":10,"docType":"asset"}, {"AppraisedValue":700,"Color":"black","ID":"asset5","Owner":"Adriana  
","Size":15,"docType":"asset"}, {"AppraisedValue":800,"Color":"white","ID":"asset6","Owner":"Michel","Size":15,"docType":"asset"}]  
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Step 10: Query a Non-Existing Asset

```
peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset50"]}'
```

```
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ peer chaincode query \  
-C mychannel \  
-n basic \  
-c '{"Args":["ReadAsset","asset50"]}'  
Error: endorsement failure during query. response: status:500 message:"The asset asset50 does not exist"  
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Step 11: Stop the Fabric Network

```
./network.sh down
```

```
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$ ./network.sh down  
Using docker and docker-compose  
Stopping network  
[+] down 10/10  
✓Container ca_org2 Removed 10.4s  
✓Container peer0.org2.example.com Removed 1.2s  
✓Container orderer.example.com Removed 0.5s  
✓Container ca_org1 Removed 10.7s  
✓Container peer0.org1.example.com Removed 1.1s  
✓Container ca_orderer Removed 10.6s  
✓Volume compose_peer0.org2.example.com Removed 0.0s  
✓Volume compose_peer0.org1.example.com Removed 0.0s  
✓Volume compose_orderer.example.com Removed 0.0s  
✓Network fabric_test Removed 0.2s  
Error response from daemon: get docker_orderer.example.com: no such volume  
Error response from daemon: get docker_peer0.org1.example.com: no such volume  
Error response from daemon: get docker_peer0.org2.example.com: no such volume  
Removing remaining containers  
Removing generated chaincode docker images  
Untagged: dev-peer0.org2.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5f6e664c87b24c035bee15cbcc7f59b629-c68f8ecc2a0ff0cc14976  
15f5e33a8bfa908e0fd3555f864cfa1398b52d341cb:latest  
Deleted: sha256:34833a4ec2367dcc4b6f0690be857cb97db0e64faa8825a1fa5e537098b6a9f  
Untagged: dev-peer0.org1.example.com-basic_1.0-1d4d445573112813889f87c307bc2b5f6e664c87b24c035bee15cbcc7f59b629-d3cf9b1f4b9747cc8433a  
c0e74bfaeade1cf0b97f5c1163835f604aa885d60dc:latest  
Deleted: sha256:78348e9dc8ce4b6a88223906ccfca466ea2750aad1c7e474381b48d2f2b4d024  
dharsna@DHARSHU:~/fabric-dev/fabric-samples/test-network$
```

Result

Ledger data was successfully queried from the Hyperledger Fabric blockchain. Existing assets were retrieved, a new asset was created and queried, the complete asset list was verified, and a non-existing asset query produced the expected error.